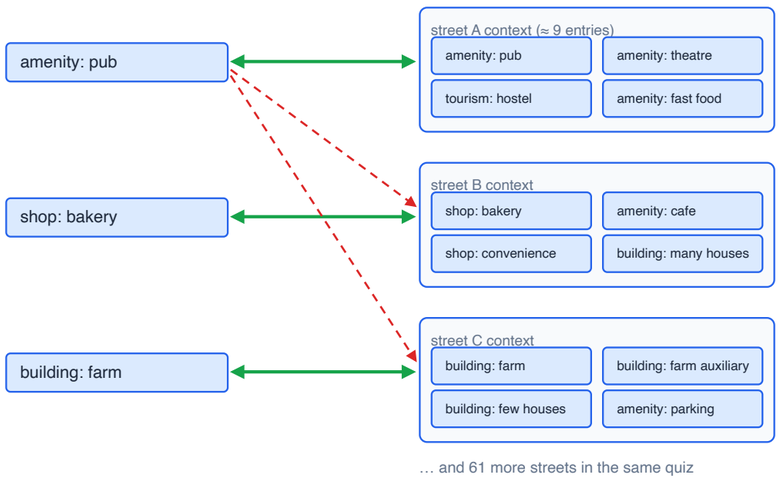
# How CityFM learns: three matching quizzes

Every stage shows the model a batch of pairs that belong together, shuffled. Training adjusts the model so that each item is more similar to its own partner than to any other item in the batch, on either side. Over thousands of quizzes, things that belong together become close together in the embedding space.

T text embedding (BERT, tags read without names) · V footprint embedding (image network + floor area) · frozen in this stage · ↔ made more similar: true partners · - - → made less similar: every other item in the batch

## 1 Stage 1 · Infomax: an entity ↔ its street

Quiz: 64 streets. One entity is picked from each street's context; it must pick out its own street's context.



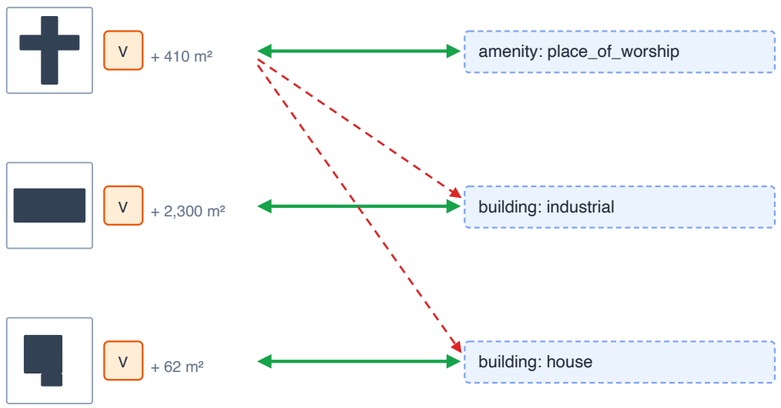

… and 61 more streets in the same quiz

| | |
|---|---|
| **Similar** | an entity and the street it is on |
| **Different** | the same entity and the other 63 streets and 63 entities |
| **Trained** | BERT + text projection (T on both sides) |
| **Should teach** | things found together become similar: pubs near theatres and hostels, farms near fields and few houses |
| **Settings (v5)** | 625 steps of 4 quizzes; names left out, so a POI is read by its type only |

**Learns.** With the POI held out, it finds the right street 7.0% of the time (top 5: 23%), against 4.2% (14%) for untuned BERT and 1.6% for guessing. Categories found together become more similar (ρ 0.62, BERT 0.18).

## 2 Stage 2 · Polygon: a footprint ↔ its building type

Quiz: 64 buildings, each with a different tag text. Each footprint (picture + floor area) must pick out its own text.



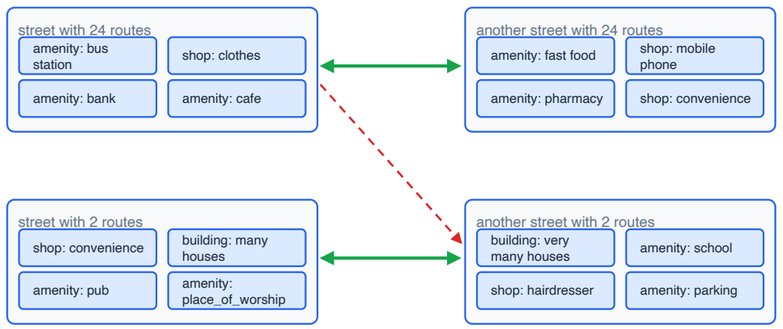

… and 61 more buildings in the same quiz

| | |
|---|---|
| **Similar** | a footprint and the text of its own building type |
| **Different** | the same footprint and the other 63 texts and 63 footprints |
| **Trained** | image network (from ImageNet) + footprint and area projections; the text side is frozen |
| **Should teach** | a footprint's embedding is most similar to the text embedding of its own type, so shape and size can predict missing tags |
| **Settings (v5)** | 1,250 steps of 4 quizzes; footprints drawn in metres; names left out (an outline cannot show a name) |

**Learns.** A building's class (18 classes) can be read from its footprint embedding 57% of the time, against 19% from floor area alone; matching footprints straight to class texts works 47% of the time (chance 6%).

## 3 Stage 3 · Relation: a street ↔ a similar street

Quiz: streets on bus routes are paired with another street carrying almost the same number of bus routes, anywhere in Leeds; each context must pick out its partner's.

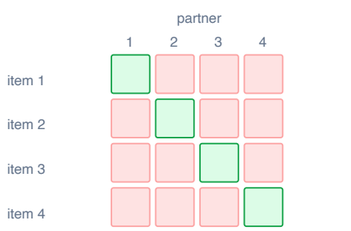



… only about 10 pairs per quiz: just 10% of streets carry a bus route

| | |
|---|---|
| **Similar** | two streets with a similar number of bus routes |
| **Different** | streets with a different number of bus routes |
| **Trained** | BERT + text projection (T on both sides), at a tenth of stage 1's learning rate |
| **Should teach** | streets with a similar role in the network get similar context embeddings |
| **Settings (v5)** | 270 steps of 1 quiz; names left out |

**Barely learns** (loss 4% of the way from guessing to the floor). Bus-route counts say little about a street's surroundings. **Planned:** match each street's context to its own network profile (integration and choice at several radii).

## What every stage does with a batch




similarity of every item to every partner (one quarter of the full 128 × 128 matrix, which also compares items with items and partners with partners)

**1.** Embed both sides and compare every item with every other item in the batch, on both sides.

**2.** The loss (InfoNCE) is low when each item's own partner (green diagonal) scores higher than all the decoys (red) and all the other items on its own side.

**3.** The weights being trained are updated to raise the green scores and lower the others, and the next batch is drawn.

**Why it works without labels:** the "right answers" come from the data itself (what is on the same street, what a building is tagged as). The decoys are the other items in the same batch, so no extra data is needed.

### What makes a good quiz

- The partner must be **predictable** from the item: names were dropped because an outline, or a pub's type, cannot predict a name.
- The batch must be **varied**: when 80% of context entries were "building: house", streets all looked alike (fixed by collapsing houses).
- The pairing must be **meaningful and cover the data**: stage 3's bus-route pairing is neither, which is why it is being replaced.